Power Electronic Converter Design

Instructions for mid-term and final seminar and reports

2024-2025

Mid-term seminar

- Date: Thu 12.12.2024, at 9-12. Teams.
- Time: **30**-40 minutes for oral presentation + **15**…20 min for questions, comments and discussion
- Format: powerpoint (pptx) recommended
- Each person in the group will present the content from the area of responsibility
    - mark who has done what

- Contents:
    - Targets of the project
    - Design process, phases, challenges, solutions
    - Simulations, models, waveforms, analysis of the simulation results
    - Reasons behind the design and choices
    - PCB design
    - Purchasing plan
    - Plan for building and testing
    - Results
    - What next…
    - Others…

Mid-term report

Deadline: check course Moodle page

· Format: pdf-file returned to Moodle. For writing use Word (docx), (LUT Thesis template for Bachelor's and Master's theses)

  - (teachers will comment the pdf-file in Moodle)
- Could be divided chapter/ group member
  - mark who has done what

· Contents:

- Targets of the project
- Design process, phases, challenges, solutions
- Simulations, models, waveforms, analysis of the simulation results
- Presentation of the results and reasoning (i.e. calculations, simulations etc.) behind the design.
- Calculations and simulation models can be added also as appendix.
- Printed circuit board (PCB) design
- Purchasing plan
  - Components, their cost, where available, delivery time estimation if possible,
- Plan for building and testing
  - for building (who will build, what and where?)
  - for testing the system (what equipment is needed for testing?)
- Reflections of the degree of difficulty and work amount of the project
  - for example, what knowledge / skills were needed to get more
  - Evaluation of working hours in different phases of the project
- Project plan included as appendix

Final seminar

- Date: Thu 17.4. 2025, at 9-11.
- Place: Laboratory 6216
    - (and Teams if not possible to be present)
- Time: 30-40 minutes for oral presentation + 15…20 min for questions, comments and discussion
- Format: powerpoint (pptx) recommended
- Each person in the group will present the content from the area of responsibility
    - mark who has done what

- Contents:
    - Briefly of the background (possible updates) and main focus on the new things: building, testing, results and analyses + self-evaluation.
        - Targets of the project
        - Design process, phases, challenges, solutions
        - Simulations, models, waveforms, analysis of the simulation results
        - Reasons behind the design and choices
        - Final purchasing list
    - **Report of building and testing**
        - use also photos
        - also video clips of building and tests are welcome
        - a promotional video of the project (optional, but has a positive effect on the grade +0.25)
            - duration max 5 min.
    - **Results and analysis**
        - also considerations how to improve the design
    - Self evaluation (see report instructions for more)
    - Others…

Final report

Deadline: check course Moodle page

- is continued and updated from mid-term report with new chapters added.
- Save As a new file and leave the original mid-term report file frozen.

· Format: pdf-file returned to Moodle. For writing use Word (docx), (LUT Thesis template for Bachelor's and Master's theses)

    - (teachers will comment the pdf-file in Moodle)
- Could be divided chapter/ group member
    - mark who has done what

· Contents:

- Targets of the project
- Design process, phases, challenges, solutions
- Simulations, models, waveforms, analysis of the simulation results
- Presentation of the results and reasoning (i.e. calculations, simulations etc.) behind the design.
- Calculations and simulation models can be added also as appendix
- Report of building and testing
    - (use also photos)
    - building (who built, what and where?)
    - testing of the system
        - (also what equipment was needed for testing)
        - measurement results
- Results and analysis
- Self evaluation (individual, separate attachment by e-mail to teacher lasse.laurila@lut.fi)
    - What was learned?
    - What was
        a) the most difficult,
        b) easiest and
        c) interesting in the project?
    - What would you do differently in the project, if the same task was started again from the beginning?
    - What grades would you give yourself and to the group on the different parts of project / (evaluation matrix)?
- Reflections of the degree of difficulty and work amount of the project
    - for example, what knowledge / skills were needed to get more
    - Evaluation of working hours per student in different phases of the project, total amount of working hours.
- Development ideas and suggestions for the course
- Appendix:
    - Project plan
    - Final purchasing list(s)
        - Components, their cost, total cost, where available, delivery time

PS. Remember to also fill the normal course feedback!

PS 2. Remember to save your files to course Teams folder